

Technical Document #635: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.
- Chunking splits documents into smaller segments for embedding.